Low Voltage Pulse-Control

Note: You can use many other microcontroller boards such as Arduino
Just pick something which is VERY predictable
may want to use a driver even with an arduino which can output 5V logic

A1
Teensy 4.1
VIN
GND
3V3
+5V
RESET
Digital_Out

U7
74HCT125N
VCC
GND
1OE
1A
2OE
2A
3OE
3A
4OE
4A
1Y
2Y
3Y
4Y

+3V3
GND
+5V

J1
J2
J3
J4
AFBR-1624Z
VCC
Ground
Data_In
Fiber_Out
TX

AFBR-2624Z
VCCR
GND
Data_Out
Fiber_In
RX

C17 10u
C18 100n
C19 10u
C20 100n
C21 10u
C22 100n
C23 10u
C24 100n
R1 4k7
R2 4k7

Note: Ideally I'd drive each gate driver individually, and read the pulse sent back from the gate-driver boards
They send short pulses at rising and falling edges to acknowledge each pulse sent to them
This can also be used to indicate Error and Fault conditions of the IGBT drivers

Isolation Barrier

IGBT Fiber-Optic Gate Drivers
Note: drew fiber lines as wires

ISO_GND
ISO_+15V

J5
J6
J7
J8
AFBR-2624Z
AFBR-1624Z
VCC
GND
Fiber_In
Data_Out
Fiber_Out
Data_In
TX
RX

U5
U6
1SD418F2-FZ1200R33KF2
Drive_signal_input
Status_Output
VCC
Ground
Emitter_Sense_Input
Gate_Output
Collector_Sense_Input

Note: Gate Drivers have Fiber Receivers and Transmitters Integrated
I just drew them this way to try to separate the fiber signals

Pulse IGBTs
In Two-Switch Forward configuration

Q1
Q2
FZ1200R33KF2
Collector_phase_1
Collector_phase_2
Collector_phase_3
Emitter_phase_1
Emitter_phase_2
Emitter_phase_3
Emitter_Sense
Gate
Collector_Sense

Note: This section of the circuit needs to handle a few thousand amp pulses
I've labeled what I used for conductors for each section
Such as 2AWG Welding Cable or 1/8Inch aluminum Buss-Bar

Snubber Capacitor Bank: each cap is 1.5uF, 3kV
These are bolted directly to the buss-bars on the IGBT modules
C3 C4 C5 C6 C7 C8 C9
C10 C11 C12 C13 C14 C15 C16
1.5µF
MES3
MES2
0-20V Voltage (+ to Supply)
0-20V Voltage (+ to Probe)
1/8 Inch Aluminum Bus-Bar
2AWG Welding Cable (Pos and Neg Twisted Together)
D1 D2 D3 D4 D5 D6
Fast-Recovery Diode
Custom TMS Coil
L1
Pulse Capacitor Bank
C1
1050µF 1.8kV
C2
1050µF 1.8kV
High-Voltage DC
Note: Connecting MES Charge-Circuit to J1 & J2
I'm using two 14AWG 300V wires for each Pos and Neg
2x 14AWG 4ft Wires

High-Voltage Charging Circuit

T1
Microwave-Oven Transformer
T2
Microwave-Oven Transformer
T3
Variable Transformer
120VAC Input
AC
High-Voltage AC

Power Supplies / DC-DC Converters

Isolated DC-DC Converter
+24V
ISO_+15V
ISO_GND
+VIN
-VIN
ON/OFF
+VOUT
-VOUT
GND

Non-Isolated DC-DC Converter
+24V
+5V
+3V3
IN
OUT
GND

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